

DS-3001

Data Analysis and Visualization

Exploratory Data Analysis

Confirmatory, Exploratory and Explanatory Data Analysis

- Confirmatory Data Analysis (CDA)
 - CDA relies on probability models (such as the t-distribution or normal distribution) where the observed data is assumed to be a random sample generated from a hypothetical population.
 - The primary goal of CDA is statistical inference. It aims to learn about an underlying, hypothetical population by evaluating hypotheses, estimating unknown parameters, calculating confidence intervals, and performing formal hypothesis tests.

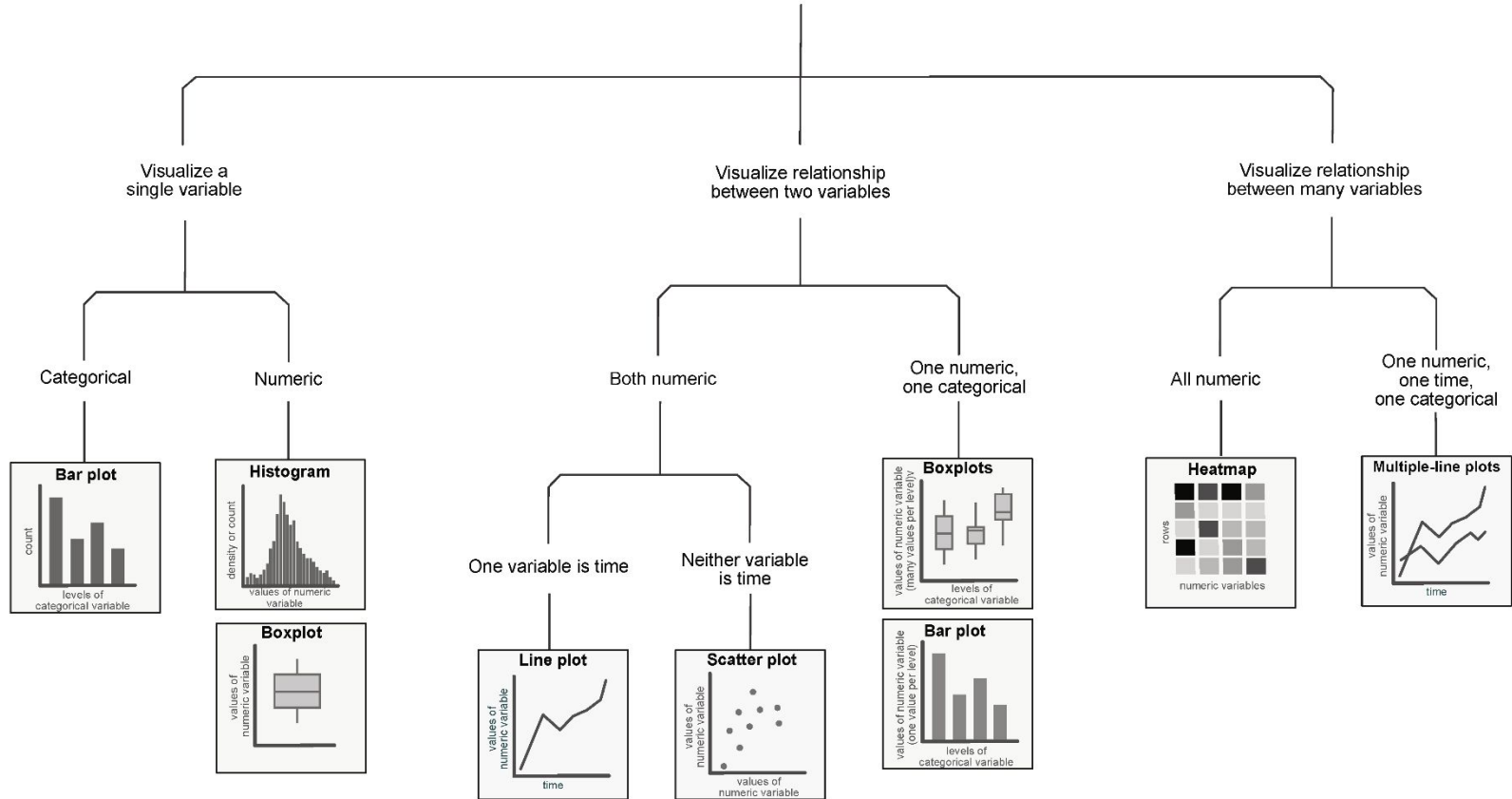
Confirmatory, Exploratory and Explanatory Data Analysis

- Exploratory Data Analysis (EDA)
 - Creating preliminary visualizations to uncover patterns and trends. Often quick, rough, and used internally by the data scientist to understand the dataset.
 - Goal: To find insights and understand the data in the context of the domain problem.
- Explanatory Data Analysis
 - Producing polished, presentation-quality figures to communicate specific findings to an external audience (team, executives, publications).
 - Goal: To tell a story or highlight specific, interesting patterns discovered during EDA.

A Question-and-Answer Workflow

- Ask a Domain Question
 - Start with specific questions related to the project goal, e.g., "Are global organ donations increasing?"
- Choose a Technique
 - Select a visualization based on the variables involved (e.g., line plot for time-series data).
- Iterate and Refine
 - Let initial answers prompt new questions, but always keep the overall project goal in focus to avoid getting lost.

What do you want to do?



References

1. Veridical Data Science by Bin Yu and Rebecca L. Barter - Chapter 5
 - <https://vdsbook.com/>